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Week 2: Quantitative-based Problems
Foundations of Machine Learning for Health and Social Care Tutorial

Wednesday 25th February 2026

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Week Two: Quantitative-based Problems

Prediction vs Estimation

Prediction: Objective is to minimise the error between observed values y and predicted values $\hat{y}$, particularly for new data points. Interpretability of individual parameters is often secondary.

Estimation: Focus is on understanding the parameters themselves. In a linear model

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \varepsilon,$$

the coefficients β_j quantify how the expected value of Y changes with a one-unit change in X_j, holding other variables constant.

These aims are related, but not identical, and often involve trade-offs.

Train and test splitting

A model that fits observed data well may still perform poorly on new data. To assess generalisation, we split the dataset into two disjoint parts.

The training set is used to estimate model parameters, while the test set is used to evaluate predictive performance:

$$\text{Training error} \neq \text{Test error}.$$

The test set must be treated as unseen data and must not influence model fitting or tuning.

K-fold cross-validation

In k-fold cross-validation, the dataset is partitioned into k equally sized folds.

For each fold $j = 1, \ldots, k$:

- the model is trained on the remaining k-1 folds,
- predictions are evaluated on fold j.

The final performance estimate is

$$\frac{1}{k}\sum_{j=1}^{k} \text{Error}_j ,$$

which reduces dependence on any single data split.

The linear regression model

In multiple linear regression, we model a continuous outcome as a linear combination of predictors:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \varepsilon_i$$

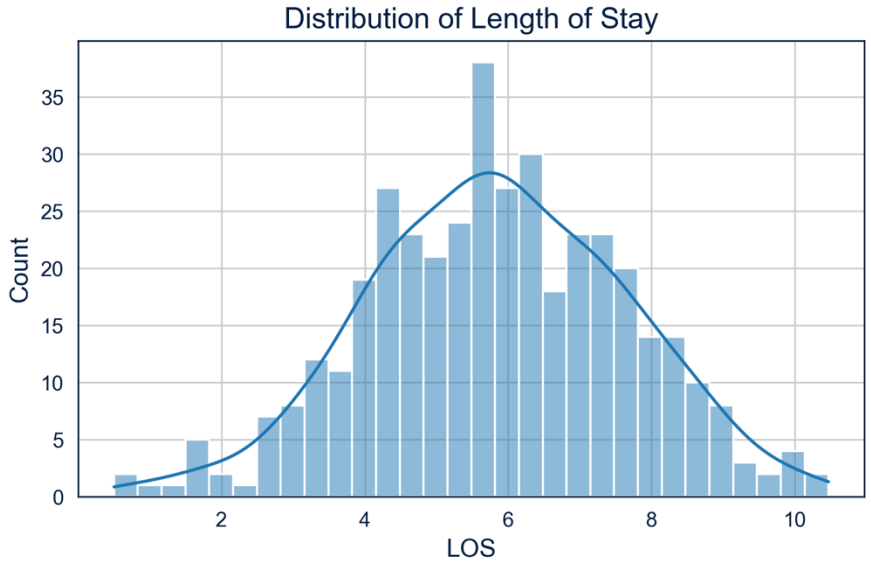
Here, the outcome y_i represents a continuous clinical measure (such as length of stay), while the predictors x_{ij} represent patient characteristics such as age or disease status.

The error term ε_i captures variation not explained by the model.

How did the model do?

Model Performance on Training and Test Data

Metric	Value
Train R^2	0.361
Test R^2	0.346
Test RMSE (days)	2.14



Variance inflation factor (VIF)

For predictor X_j, the VIF is defined as:

$$\text{VIF} = \frac{1}{1 - R_j^2}$$

where R_j^2 is the coefficient of determination obtained by regressing X_j on all the other predictors.

Variance Inflation Factor (VIF) for Predictors

Variable	VIF
Age	1.6567657015417603
ChronicIllness	1.6567657015417603

The core idea of polynomial regression

Polynomial regression extends linear regression by allowing predictors to appear as powers.

Instead of modelling the outcome as a straight line in a predictor x, we allow curved relationships by including terms such as x^2, x^3, and so on:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \cdots + \beta_d x^d + \varepsilon$$

Although the relationship between x and y is non-linear, the model remains linear in the parameters, which is crucial.

Elastic net: Combining ridge and lasso

Elastic Net combines both penalties:

$$\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda[\alpha \sum_{j=1}^{p}|\beta_j| + (1 - \alpha)\sum_{j=1}^{p}\beta_j^2]$$

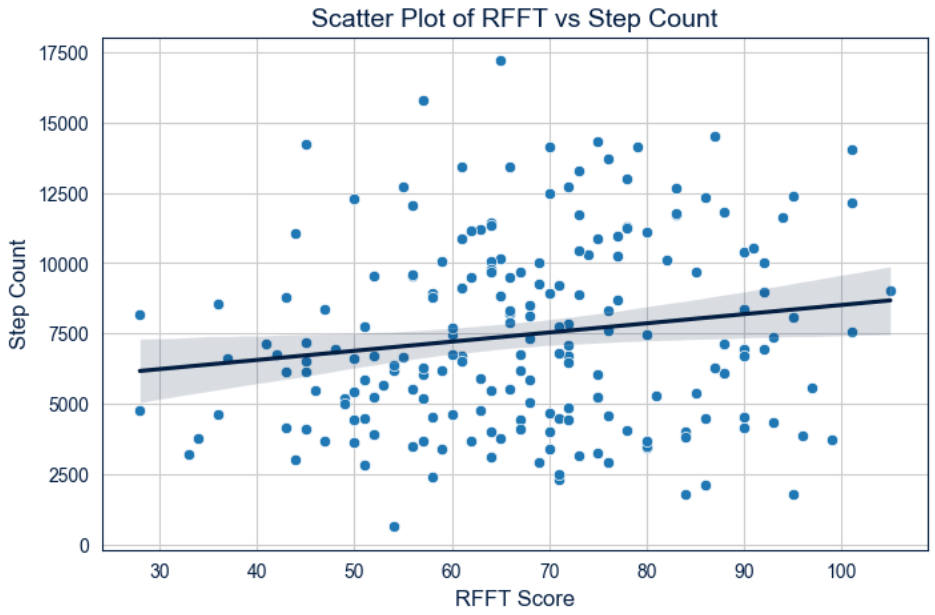
Here:
λ controls overall regularisation strength,
$\alpha \in [0,1]$ controls the balance between Lasso and Ridge.

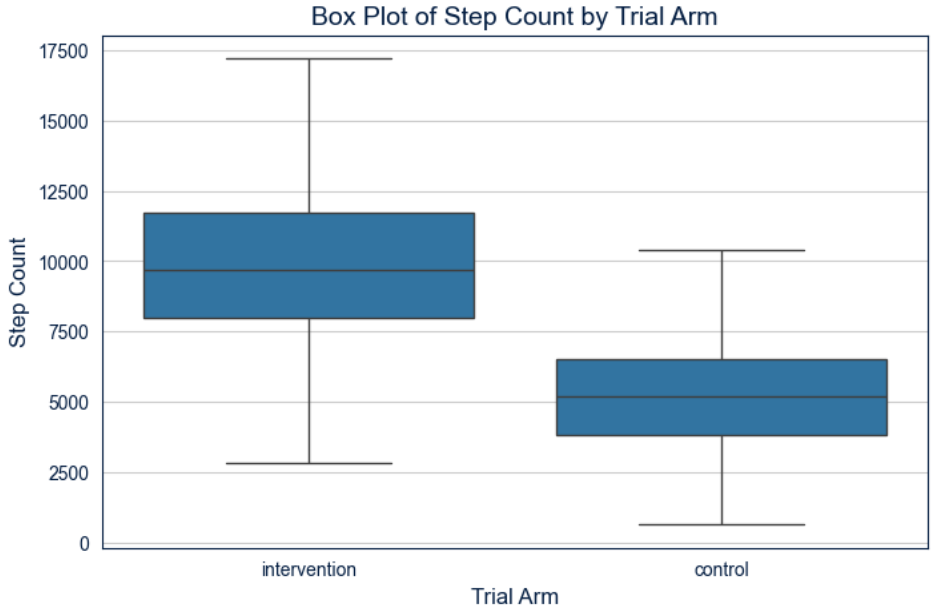
Any Questions?



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Paired Programming Practice





- Using data visualization to explore the relationship between predictors and outcomes.
- Fit and compare multiple different regression models.
- Evaluate model performance.